Federal Research and Development Tax Credit Study

McGrath Electric, Inc

2023

1) Executive Summary

McGrath Electric, Inc ("the Company") engaged Occams Advisory ("Occams") to perform a Research and Development ("R&D") Tax Credit Study ("Study"). The Study was undertaken to analyze and document expenditures and business activities and determine qualified research expenses that may be applied towards the tax credit following the guidance of Internal Revenue Code ("IRC" or the "Code") Sections 41 and 174.

The Study covered the activities undertaken and the expenses incurred by McGrath Electric, Inc during the tax year(s) 2023 (the "Study Period"). The results of the Study yielded $67,794.40 in Federal R&D tax credits.

In order to analyze, substantiate and document the Company's various R&D activities in accordance with the Code, associated regulations, and various case law, the Company collected the following information:

- General demographic information related to the Company's formation and qualification for the R&D tax credit;
- Federal and State (if applicable) tax return information;
- Employee list with job titles for the Study Period;
- Employee payroll amounts (W-2 box 1 wage equivalent) for each employee for the Study Period;
- A listing of qualified supplies, contractors, and cloud computing expenses (if applicable);
- A listing of projects completed during the Study Period;
- Time sheet data and/or estimates of qualified hours and activities; and
- In-depth project information related to qualified research activities.

Following the collection of the above information, Occams and/or its agents analyzed the data to ensure that the work completed, along with the associated expenditures qualified for the R&D tax credit by meeting the four-part-test outlined in the Code. Furthermore, Occams and/or its agents analyzed the scope of work and/or associated contractual agreements to determine if any projects would fall under the funded research exclusion of the Code or any other associated exclusions.

a) Summary of R&D Expenditures

	Tax Year 2023
Qualified Wages	$834,527.07
Total Qualified Research Expenses	$834,527.07
Federal R&D Tax Credit	$67,794.40

b) Summary of Representative R&D Projects

A summary of the projects analyzed during the Study along with their qualifying status is contained in the table below.

Project Name	Qualification Status
General Development Activities	Qualified

2) Overview of Statutory Authority

a) Qualified Research

IRC § 41 provides a tax credit for conducting research activities undertaken to develop or improve a business component. Research constitutes "qualified research" under § 41(d)(1) if it is research:

(1) With respect to which expenditures may be treated as expenses under §174;

(2) That is undertaken for the purpose of discovering information that is technological in nature;

The application of which is intended to be useful in the development of a new or improved business component of the taxpayer; and

(3) Substantially all of the activities of which constitute elements of a process of experimentation that relates to a qualified purpose.

Under §174 qualified expenses are amounts paid or incurred by the taxpayer in the taxable year in connection with his trade or business as expenses which are applied towards research or experimental activity. Expenses qualify as research or experimental costs when they are incurred in the effort to resolve uncertainty in the development or improvement of a business component. Uncertainty exists if the information available to the taxpayer does not establish the capability or method for developing or improving the business component, or the appropriate design for the business component.

To satisfy the second requirement of the statute, the research must be undertaken to discover information rooted in the principles of science. This component does not require a taxpayer seeking to obtain information that exceeds, expands or refines the common knowledge of skilled professionals in a particular field of science; nor does the component require a taxpayer to succeed in the research.

The third requirement is that the application of the research be applied to the development of a new or improved business component of the taxpayer. A business component is a product, process, formula, invention, software, or technique produced in the scope of the taxpayer's trade.

Regarding the last requirement, substantially all of a taxpayer's research activities will be considered qualified research if 80 percent or more of the research activities constitute elements of a process of experimentation that relates to a qualified purpose. A process of experimentation is a process designed to evaluate one or more alternatives to achieve a result where the capability or the method of achieving that result, or the appropriate design of that result, is uncertain as of the beginning of

the taxpayer's research activities. A process of experimentation must fundamentally rely on the principles of science and involve the identification of uncertainty concerning development or improvement of a business component, the identification of one or more alternatives intended to eliminate that uncertainty, and the identification and the conduct of a process of evaluating alternatives (through, for example, modeling, simulation, or a systematic process of trial and error). This process of experimentation is undertaken for a qualified purpose if it relates to a new or improved function, performance, reliability or quality of the business component.

b) Excluded Activities

IRC § 41(d) (4) outlines activities for which the tax credit is not allowed. The excluded activities that do not meet the definition of "qualified research" are as follows:

(1) Any research conducted after the beginning of commercial production of the business component.

(2) Any research related to the adaptation of an existing business component to a particular customer's requirement or need.

(3) Any research related to the reproduction of an existing business component (in whole or in part) from a physical examination of the business component itself or from plans, blueprints, detailed specifications, or publicly available information with respect to such business component.

(4) Any –

(a) efficiency survey,

(b) activity relating to management function or technique,

(c) marketing research, testing, or development (including advertising or promotions),

(d) routine data collection, or

(e) routine or ordinary testing or inspection for quality control.

(5) Except to the extent provided in regulations, any research with respect to computer software which is developed by (or for the benefit of) the taxpayer primarily for internal use by the taxpayer, other than for use in –

(a) an activity which constitutes qualified research (determined with regard to this subparagraph), or

(b) a production process with respect to which the requirements of paragraph (1) are met.

(6) Any research conducted outside of the United States, the Commonwealth of Puerto Rico, or any possession of the United States.

(7) Any research in the social sciences, arts, or humanities.

(8) Any research to the extent funded by any grant, contract, or otherwise by another person (or governmental entity).

c) Qualified Research Expenses

IRC § 41(b)(1) explains that the term "qualified research expenses" means the sum of in-house research expenses and contract research expenses. Additionally, IRC § 41(b)(2)(A) further defines in-house research expenses as:

(1) Any wages paid or incurred to an employee for qualified services performed by such employee; and

(2) Any amount paid or incurred for supplies used in the conduct of qualified research.

Furthermore, IRC § 41(b)(3)(A) defines contract research expenses as 65 percent of any amount

paid or incurred by the taxpayer to any person (other than an employee of the taxpayer) for qualified research.

i) Wages

Qualified wages are those amounts paid or incurred to an employee for qualified services performed by such employee. IRC § 41(b)(2)(B) provides that the term "qualified services" means services consisting of – 1) engaging in qualified research, or 2) engaging in the direct supervision or direct support of research activities which constitute qualified research. Furthermore, if substantially all of the services performed by an individual for the taxpayer during the taxable year consists of services meeting the requirements of clause 1 or 2, the term "qualified services" means all of the services performed by such an individual for the taxpayer during the taxable year.

Treasury Regulations § 1.41-2(d) expands upon the topic of qualified wages with the following:

> Wages paid to or incurred for an employee constitute in-house research expenses only to the extent the wages were paid or incurred for qualified services performed by the employee. If an employee has performed both qualified services and nonqualified services, only the amount of wages allocated to the performance of qualified services constitutes an in- house research expense. In the absence of another method of allocation that the taxpayer can demonstrate to be more appropriate, the amount of in-house research expense shall be determined by multiplying the total amount of wages paid or incurred for the employee during the taxable year by the ratio of the total time actually spent by the employee in the performance of qualified services for the taxpayer to the total time spent by the employee in the performance of all services for the taxpayer during the taxable year.

ii) Supplies

IRC § 41(b)(2)(B) defines "supplies" as any tangible property other than 1) land or improvements to land, and 2) property of a character subject to the allowance for depreciation. Clause 2 is not satisfied unless the property is depreciable in the hands of the taxpayer (See TG Missouri Corp. v. Commissioner, 133 TC 278, 292 (2009)).

Supply expenses are properly claimed when used in the conduct of qualified research by the taxpayer. Treasury Regulations § 1.41-2(b) (1) explains that expenditures for supplies or for the use of personal property that are indirect research expenditures or general and administrative expenses do not qualify as in- house research expenses.

iii) Contract Research Expenses

IRC § 41(b)(3)(A) defines "contract research expenses" as 65 percent of any amount paid or incurred by the taxpayer to any person (other than an employee of the taxpayer for qualified research). Essentially, contract research expenses are amounts expended for qualified services performed by a third party.

3) R&D Tax Credit Analysis by Project

i) Basic data

Project Name: General Development Activities

Project Years: 2023

Contract Type: Lump sum/fixed fee

Employees Performing Research: Adell McGrath, Andres R Duenas, Andrew A Hartmann, Andy A Rusk, Armando V Munoz, Brandon J Hatter, Brandon Stewart, Brent W Smith, Cesar N Sandoval, Daniel Nevarez, Danny A Frandsen, David A Zacarias, Eric McGrath, Erik M Core, Gavin M McGrath, Gino E Gonzales, Griffin D McGrath, Hayden C Gregonis, Heysel C Smith, Jeffrey Chapman, Jenine A Wixom, Jessica D Marmolejo, Juan J Vazquez, Justin Benzler, Kyle H Tuck, Manuel Maciel, Nick F Luczak, Stacy L Wallace, Steven E Jelich, Steven E Matthews, Timothy D Cottril, Torre A Parker Jr., and Ulises A Aldape

ii) Project Description

The company is generally in the business of electrical design and build construction.

As part of the company's business strategy, its research activities generally involve evaluating alternative designs and more efficient constructability methods.

The company focuses on using engineering and electrical engineering to create a new design or improve an existing design to be more functional and cost-efficient.

iii) Why This is a New or Improved Business Component

This project introduces novel or enhanced business components targeting performance, functionality, quality, or reliability in the following ways:

1) Resolving Design and Construction Uncertainties through CAD Modeling and On-Site Testing

- **Process:** The activity of resolving design and construction uncertainties through CAD modeling and on-site testing represents a new or improved business component targeting performance, functionality, quality, and reliability. By systematically running calculations, modeling designs in a CAD system, and conducting on-site testing, the company enhances the accuracy and feasibility of its electrical design and build projects. This approach ensures that potential interferences are identified and mitigated, leading to more efficient and reliable construction outcomes.
- **Technique:** The activity of resolving design and construction uncertainties through CAD modeling and on-site testing represents a new or improved business component targeting performance, functionality, and reliability. By utilizing advanced CAD systems to model designs, run calculations, and simulate environments, the company enhances the accuracy and efficiency of its designs. Additionally, on-site testing ensures that any unanticipated issues are identified and resolved, improving the overall reliability and functionality of the construction projects.

iv) Why This Project Involves Elimination of Uncertainty (Information Sought to Discover)

There is uncertainty regarding the capability, method, or design of the following technical challenges.

1) Resolving Design and Construction Uncertainties through CAD Modeling and On-Site Testing

Appropriate Design: One major uncertainty is ensuring that the initial design is fully comprehensive and accounts for all potential issues that may arise during the construction phase. Engineers often find themselves having to modify designs to address unexpected technical challenges that were not anticipated during the initial design phase. This ongoing risk of unforeseen issues requires a robust process of design validation and adaptation.

Method: The method of resolving design uncertainties through both CAD modeling and on-site testing introduces uncertainties regarding the accuracy and reliability of the simulations. Engineers must rely on these methods to identify potential problems, but the accuracy of the simulations can vary, and some issues may only become apparent during actual construction. This dual-method approach requires careful coordination and iterative testing to ensure that all design aspects are thoroughly vetted.

v) Why This Project Involves a Process of Experimentation

1) Resolving Design and Construction Uncertainties through CAD Modeling and On-Site Testing

The General Development Activities undertaken by the company involved significant technical uncertainties, particularly related to the design and construction phases. During the design phase, the engineers faced uncertainties regarding the feasibility and efficiency of alternative designs. To address these uncertainties, the team employed a process of experimentation that included running detailed calculations and using CAD modeling to simulate the design. This approach allowed them to systematically evaluate different design alternatives, identify potential issues, and make necessary adjustments before moving forward with the construction phase. The use of CAD modeling enabled the engineers to visualize the designs in a virtual environment and test them under simulated conditions, thereby reducing the risk of encountering unforeseen issues during actual construction.

One specific uncertainty encountered during the construction phase was the potential for unexpected technical challenges that were not accounted for during the design phase. To resolve these uncertainties, the company implemented a robust process of on-site testing. Once the design was completed in the CAD system, the engineers conducted on-site checks to ensure that there were no interferences that the CAD program might have missed. This involved a systematic trial and error approach, where the engineers would physically inspect the construction site, compare it with the CAD models, and make necessary adjustments to the design. This iterative process of design and evaluation ensured that the final construction was both feasible and efficient, addressing any technical uncertainties that arose during the construction phase.

Another area of uncertainty was the overall constructability and efficiency of the design. The company's engineers used simulations and modeling to test various construction methods and sequences to determine the most efficient and effective approach. By creating multiple design iterations and analyzing their performance through simulations, the team was able to identify and implement the best constructability methods. This process of experimentation, which involved continuous feedback and iterative improvements, allowed the engineers to refine their designs and construction plans, ensuring that the project met the desired functional goals.

The engineers relied on their expertise to conduct detailed analyses, run simulations, and perform on-site tests. This technological approach ensured that the research activities were both systematic and evaluative, focusing on eliminating uncertainties related to the design and construction of the building. By leveraging advanced engineering techniques and tools, the company was able to achieve new or improved designs that were more functional, ultimately meeting the project's objectives.

vi) Why This Project Is Technological In Nature

1) Resolving Design and Construction Uncertainties through CAD Modeling and On-Site Testing

Engineering:

The activity 'Resolving Design and Construction Uncertainties through CAD Modeling and On-Site Testing' applied the principle of engineering by addressing technical uncertainties that arise during both the design and construction phases of projects. Engineering principles were utilized to systematically identify, analyze, and resolve these uncertainties to ensure the functionality and efficiency of the designs.

Engineering's relevance to this activity is evident in the use of CAD (Computer-Aided Design) modeling to simulate and test various design scenarios. The engineers employed CAD systems to run calculations and model the design to determine its feasibility. This involved checking for interferences and assessing the design under simulated environments, which are key aspects of engineering problem-solving and design validation.

The practical application of engineering principles continued with on-site testing once the design phase was completed. Engineers conducted on-site evaluations to verify that the CAD models accurately represented the physical conditions and to identify any unforeseen interferences. This step ensured that any discrepancies between the modeled and actual conditions were addressed promptly, thereby enhancing the reliability and accuracy of the final construction.

The systematic approach to resolving uncertainties through CAD modeling and on-site testing exemplifies the core engineering principles of design, analysis, and optimization. By leveraging these principles, the

company was able to create more functional designs, ultimately contributing to the successful completion of their electrical design and build construction projects.

Mathematics:

The activity of resolving design and construction uncertainties through CAD modeling and on-site testing applied the principle of mathematics extensively throughout its execution. Mathematics played a fundamental role in running calculations and modeling out the design in the CAD system to determine the plausibility of the design. These calculations involved precise measurements and mathematical algorithms to ensure that the design met the necessary specifications and would function as intended in the real world.

Mathematics was also pivotal in the system's ability to check for interferences. By using mathematical models and simulations, the CAD system could identify potential conflicts or issues within the design that might not be immediately apparent. This process relied on mathematical principles to analyze spatial relationships and geometries, ensuring that all components of the design fit together seamlessly without any conflicts.

Once the design was completed in the CAD system, the engineers conducted on-site testing to verify the design's accuracy. This step involved additional mathematical calculations to compare the on-site conditions with the modeled design. Any discrepancies found during this phase required further mathematical analysis to adjust and refine the design, ensuring it met all the necessary criteria and functioned correctly in the actual construction environment.

The relevance of mathematics to this activity is evident in the systematic approach taken to resolve uncertainties at every stage of the project. From initial design calculations to on-site verification, mathematical principles provided the necessary framework for analyzing data, running simulations, and making informed decisions. This rigorous application of mathematics ensured that the final design was functional, aligning with the company's goals of improving constructability methods and creating more efficient designs.